LUKAS POPP

PORTFOLIO

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Bachelor Thesis

Topic: Creation of a web application based on WebGL and Three.js that let's you change the colorway and material of a shoe online.

Technologies: The shoe was rendered via PBR with deferred rendering in a from-scratch WebGL shader to fit the visual standards of adidas where the thesis was written at.

Programming languages: JavaScript, GLSL

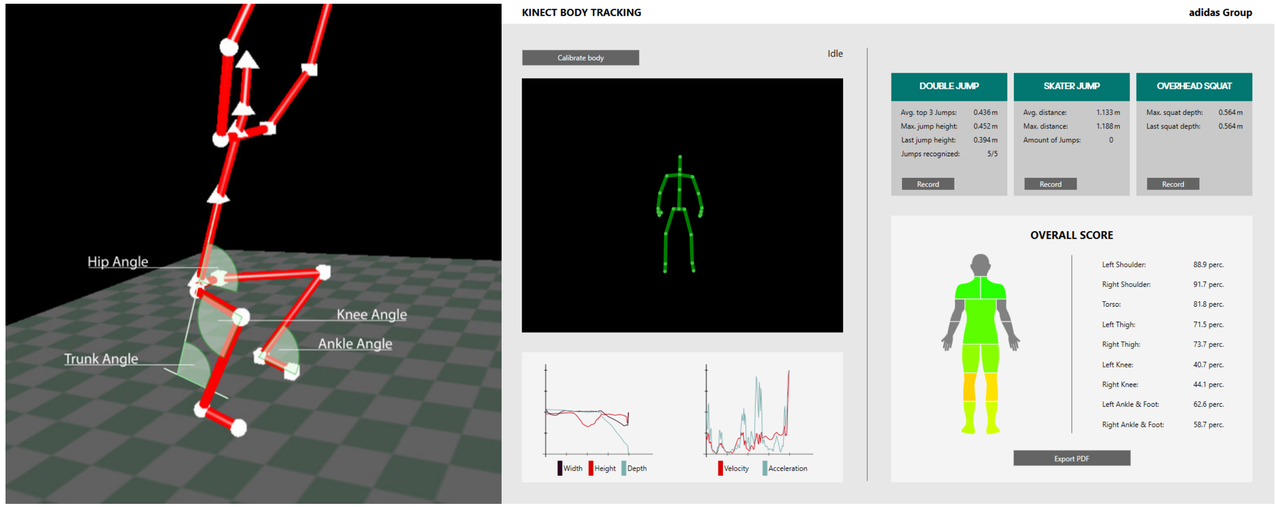
The via PBR rendered shoe with varying materials under different lighting in two environments.

Work at adidas

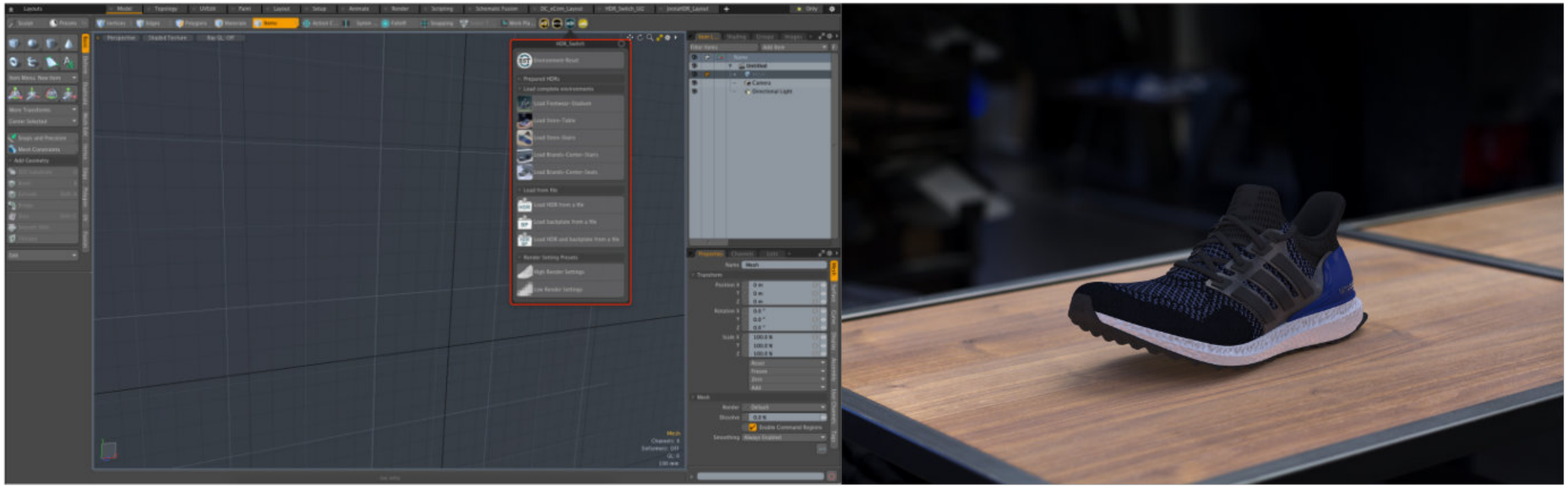
Tasks:

- Creation of an applications that detects body movements in a workout using the Microsoft Kinect as a motion detection sensor
- Development of multiple plugins for the 3D modeling software modo and Cinema 4D

Programming languages: C++, Python



Top: The movement detection application.
Bottom: An example of a rendering that was created with one of the programmed plugins.

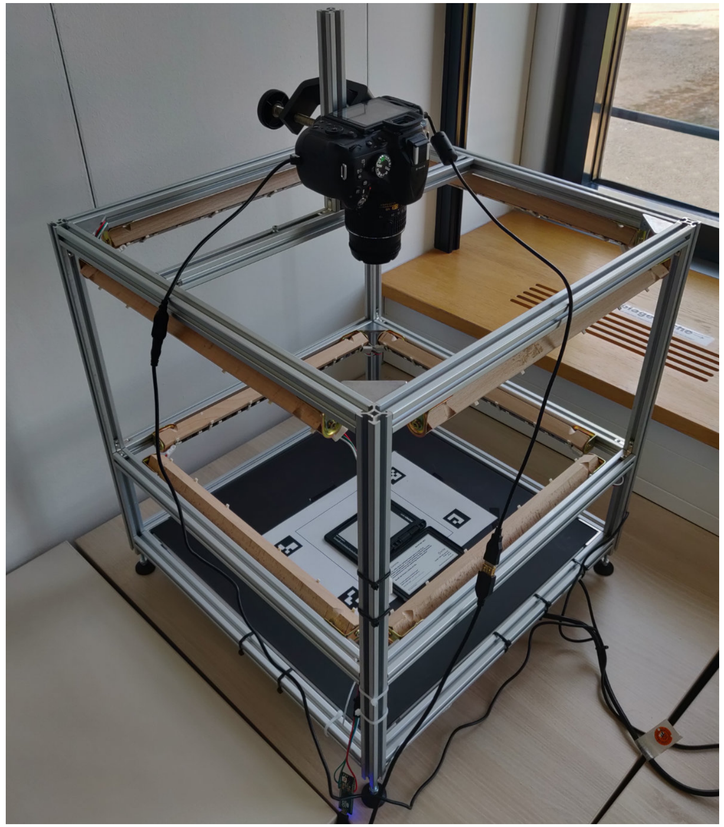


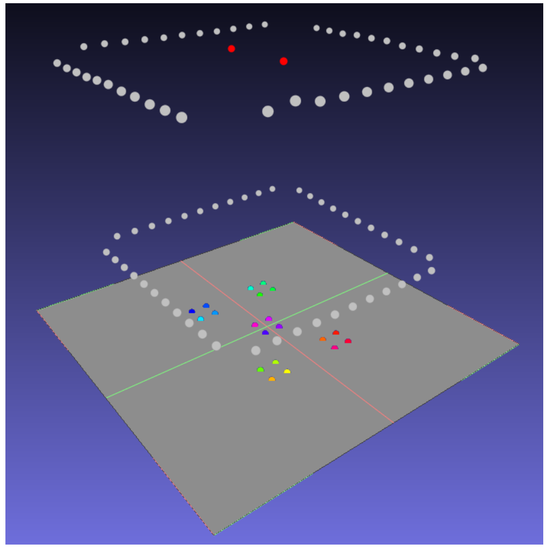
Master Project

Topic: Analyzing images of a surface recorded in a lightstage to estimate surface properties. A big part of the project was calibrating the light contribution from the LEDs in the lightstage.

Approach: I used image recognition techniques to detect hightlights in the image to calcualte the light positions. This was possible through ArUco markers on the bottom and inserting a mirror into the frame.

Programming languages: C++ with OpenCV





The lightstage at the top and to the right the calculated LED, the marker and the camera positions.

Master Thesis

Topic: Finding and testing a new way to create and display Reflectance Transformation Imaging (RTI) files of paintings. RTIs allow to relight surfaces in a shader retrospectively to highlights surface details.

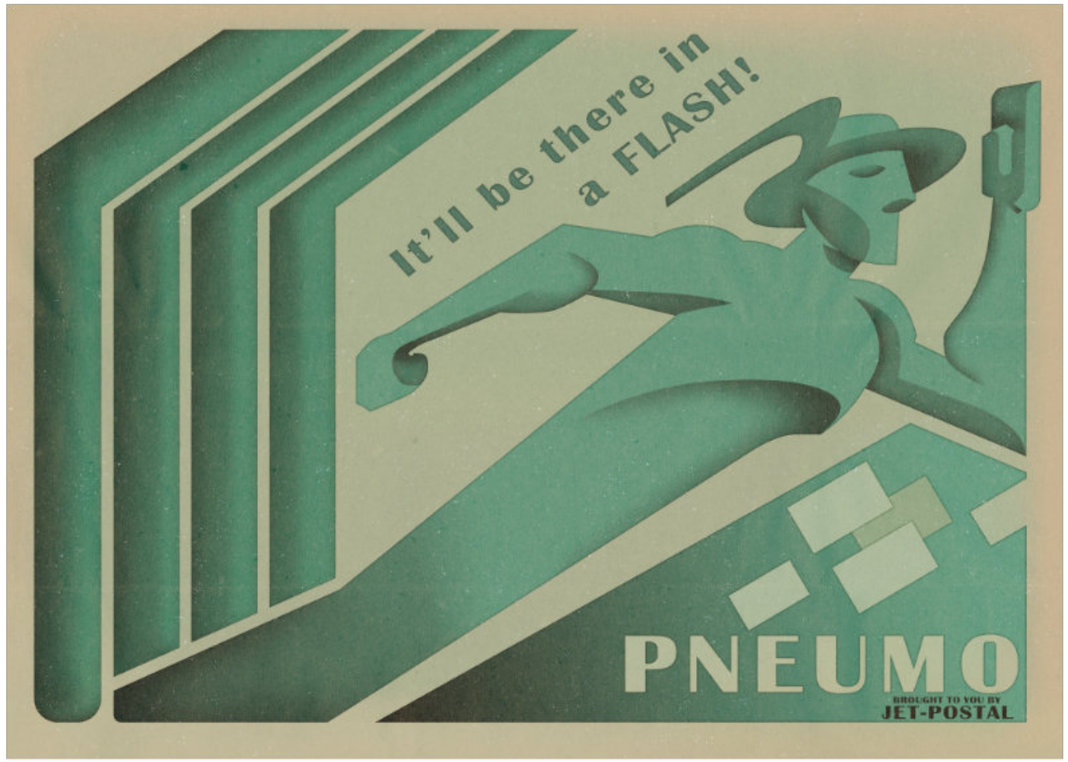
Approach: Paintings were recorded and lit from various angles with a smartphone while being recorded simultaneously with a static camera. Through the video, the camera position was estimated in every frame and synchronized with the static camera. An RTI file was created from multiple frames.

Programming languages: C++, GLSL

Two paintings recreated and rendered with the developed tool.

Creative Work

I'm vividly interested in design in general. For example, out of own interest, I recreated posters from a video game in my own style.



This is just a small overview over some visually interesting projects I worked on. If you're interested in more of my work, you can contact me at luk.popp0@gmail.com.